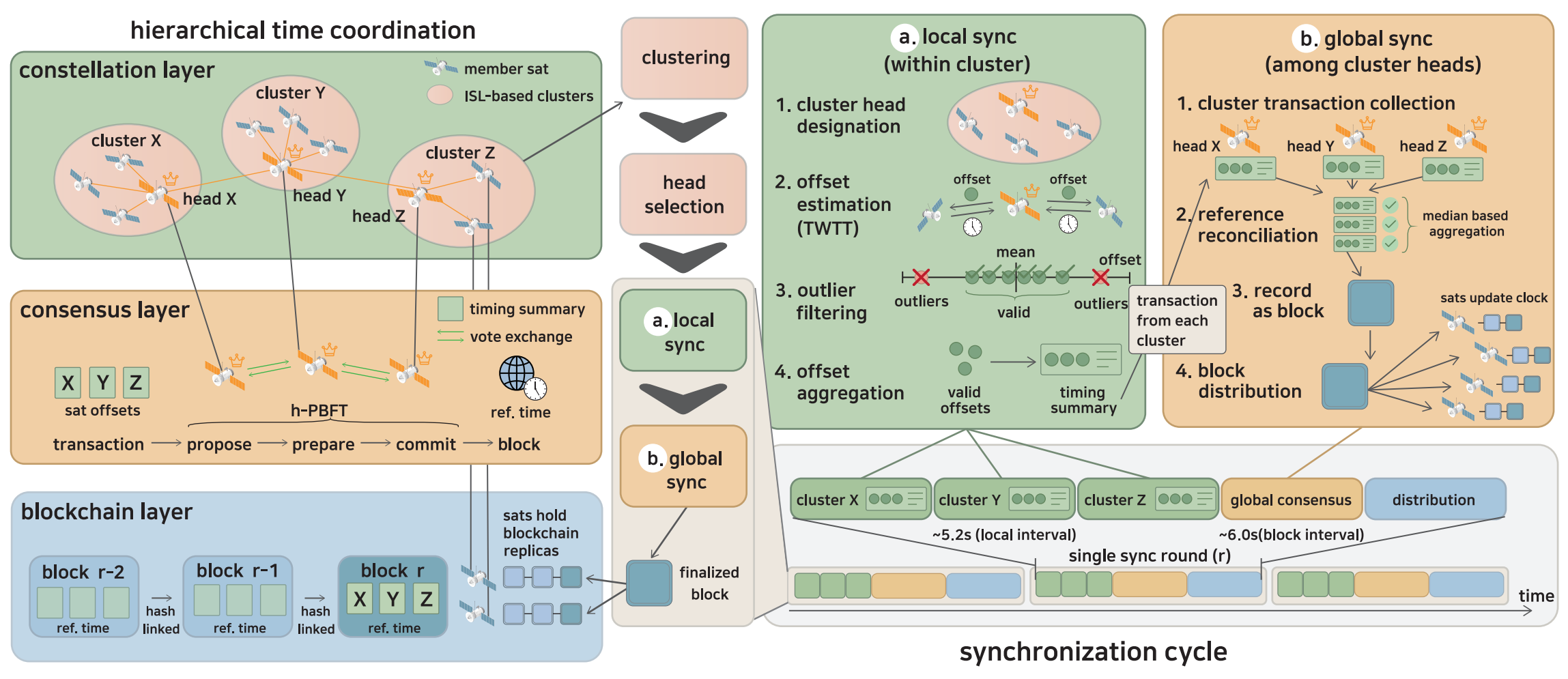

hierarchical time coordination
constellation layer
member sat
ISL-based clusters
cluster X
cluster Y
cluster Z
head X
head Y
head Z
consensus layer
timing summary
vote exchange
X Y Z
sat offsets
h-PBFT
ref. time
transaction
propose
prepare
commit
block
blockchain layer
sats hold blockchain replicas
block r-2
block r-1
block r
X Y Z
ref. time
hash linked
clustering
head selection
a. local sync
b. global sync
finalized block
a. local sync (within cluster)
1. cluster head designation
2. offset estimation (TWTT)
offset
offset
3. outlier filtering
mean
offset
outliers
valid
outliers
4. offset aggregation
valid offsets
timing summary
transaction from each cluster
b. global sync (among cluster heads)
1. cluster transaction collection
head X
head Y
head Z
2. reference reconciliation
median based aggregation
3. record as block
sats update clock
4. block distribution
cluster X
cluster Y
cluster Z
global consensus
distribution
~5.2s (local interval)
~6.0s(block interval)
single sync round (r)
time
synchronization cycle